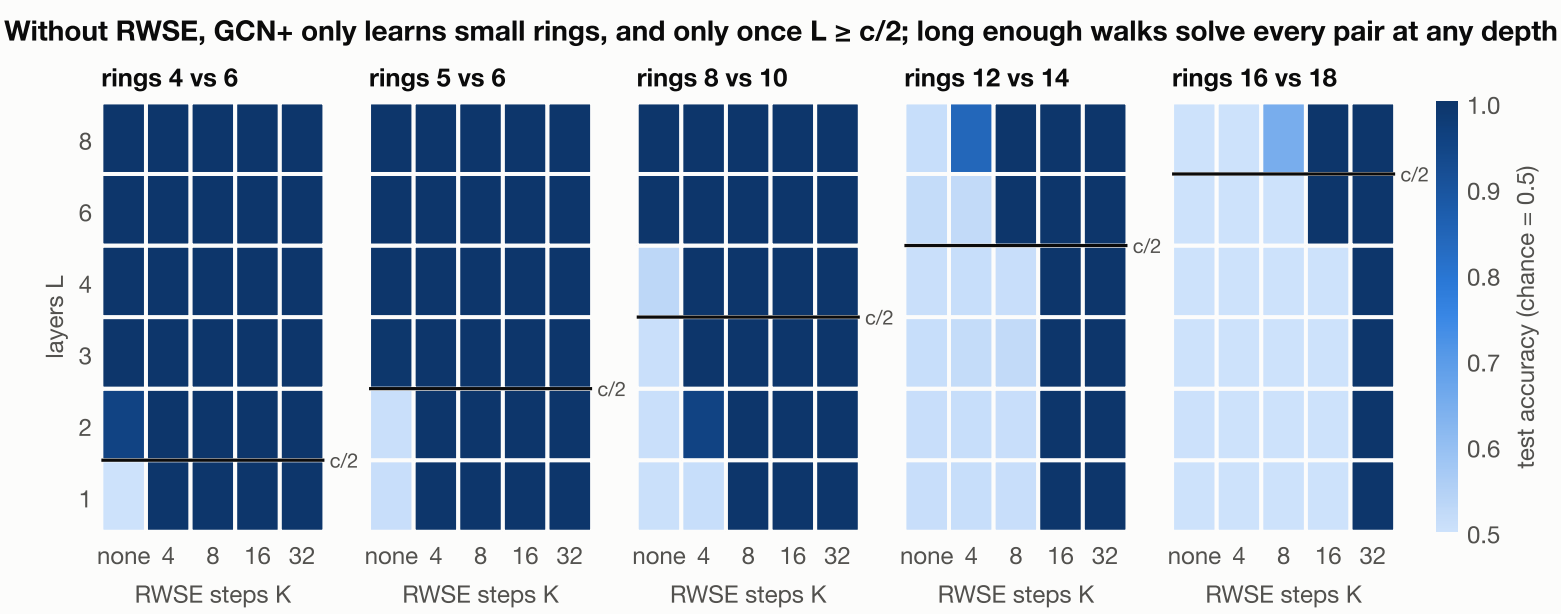
Without RWSE, GCN+ only learns small rings, and only once L ≥ c/2; long enough walks solve every pair at any depth
rings 4 vs 6
rings 5 vs 6
rings 8 vs 10
rings 12 vs 14
rings 16 vs 18
layers L
8
6
4
3
2
1
c/2
none 4 8 16 32
RWSE steps K
test accuracy (chance = 0.5)
1.0
0.9
0.8
0.7
0.6
0.5